

Minutes of Project Meetings: PRML Project

Meeting 1

Date: March 2, 2025

Time: 9:15 PM - 11:15 PM

Venue: Shamiyana

Attendees: Sonic Vyas, Shlok Kanani, Shardul Vikram Singh, Sarthak Bhiwaji Kolekar, Yash Kumar Singh

Discussion Points:

- Finalized broad project idea: Recognizing hand-drawn sketches using kNN and SVM.
- Defined individual roles based on expertise.
- Planned initial dataset exploration and preprocessing using HOG and LBP feature extraction.

Action Items:

- Shardul to organize the dataset and define evaluation metrics.
- Shlok to set up preprocessing pipeline and kNN model.
- Sonic to begin SVM model implementation and feature extraction.
- Sarthak to review preprocessing outputs and assist with debugging.
- Yash to draft an initial project report introduction.

Written by: Yash Kumar Singh

Meeting 2

Date: March 9, 2025

Time: 10:00 PM - 12:00 AM

Venue: Sarthak's Room

Attendees: All members

Discussion Points:

- Reviewed dataset organization and preprocessing status.
- Assessed initial SVM model results and feature extraction.

- Discussed kNN model setup and different values of k.

Action Items:

- Shardul to validate dataset split and finalize evaluation metrics.
- Shlok to refine kNN preprocessing and experiment with different k values.
- Sonic to enhance feature extraction and implement StandardScaler for SVM.
- Sarthak to tune initial hyperparameters for both models.
- Yash to update the documentation with progress details.

Written by: Yash Kumar Singh

Meeting 3

Date: March 16, 2025

Time: 9:45 PM - 11:45 PM

Venue: Sarthak's Room

Attendees: All members

Discussion Points:

- Reviewed kNN model results and comparisons with SVM.
- Identified areas of improvement in feature selection (HOG vs LBP).
- Discussed hyperparameter tuning methods for SVM kernel selection.

Action Items:

- Shardul to analyze and report evaluation metric trends.
- Shlok to refine kNN feature selection and preprocessing pipeline.
- Sonic to optimize SVM kernel selection and tuning.
- Sarthak to conduct hyperparameter tuning experiments for both models.
- Yash to document findings and prepare mid-project report draft.

Written by: Yash Kumar Singh

Meeting 4

Date: March 23, 2025

Time: 10:30 PM - 12:30 AM

Venue: Sonic's Room

Attendees: All members

Discussion Points:

- Compared final results of SVM and kNN models on test dataset.
- Evaluated performance metrics and identified key insights.
- Discussed potential improvements in feature engineering and classifier tuning.

Action Items:

- Shardul to summarize evaluation insights for both models.
- Shlok to document preprocessing methodology and feature extraction techniques.
- Sonic to test alternative feature extraction techniques (e.g., different HOG parameters).
- Sarthak to debug final implementation issues and hyperparameter tuning.
- Yash to refine project report for mid-term evaluation.

Written by: Yash Kumar Singh

Meeting 5

Date: March 29, 2025

Time: 9:20 PM - 11:20 PM

Venue: Shardul's Room

Attendees: All members

Discussion Points:

- Finalized mid-term evaluation report and presentation.
- Addressed last-minute debugging and formatting of results.
- Planned presentation delivery and role division.

Action Items:

- Shardul to finalize evaluation summary for SVM and kNN.
- Shlok & Sonic to ensure reproducibility of models.
- Sarthak to cross-check hyperparameter tuning outputs and debugging.
- Yash to compile and proofread the final report and slides.

Written by: Yash Kumar Singh

Meeting 6

Date: April 6, 2025

Time: 10:00 PM – 12:00 AM

Venue: Shlok's Room

Attendees: All members

Discussion Points:

- Finalized implementation of CNN-based feature extractor.
- Evaluated accuracy of hybrid models: CNN+KNN (44.7%), XGBoost+CNN (44.5%), and Random Forest (39.1%).
- Discussed feature importance: CNN middle-layer activations and HSV color histograms found most relevant.
- Reviewed preliminary ensemble strategy using accuracy-diversity trade-off.

Action Items:

- Shlok to update CNN+KNN results and integrate into report.
- Sonic to finalize XGBoost+CNN pipeline and verify reproducibility.
- Shardul to compile feature importance visualizations.
- Sarthak to start ensemble implementation.
- Yash to update comparative analysis in report.

Written by: Yash Kumar Singh

Meeting 7

Date: April 10, 2025

Time: 9:30 PM – 11:30 PM

Venue: Discord Voice Call

Attendees: All members

Discussion Points:

- Completed first working version of ensemble model (CNN+KNN, XGBoost+CNN, Random Forest).
- Discussed weighted voting mechanism with class-specific precision-based weights.
- Reviewed GitHub commits (April 9–10): Added ensemble scripts and updated README; project report structure updated.

Action Items:

- Shlok to conduct validation using calibrated ensemble confidence scores.
- Sonic to fine-tune voting weights (alpha parameter).
- Shardul to benchmark final ensemble accuracy and top-5 score.
- Sarthak to check for performance improvements across difficult classes.
- Yash to revise ensemble methodology section of report.

Written by: Yash Kumar Singh

Meeting 8

Date: April 12, 2025

Time: 10:00 PM – 11:45 PM

Venue: Sonic's Room

Attendees: All members

Discussion Points:

- Analyzed confusion matrices for ensemble vs. individual models.
- Reviewed common misclassification cases and complementary model strengths.
- Conducted final run of ensemble model achieving 47.9% accuracy (76.4% top-5).
- Evaluated model performance vs. resource trade-offs (inference time, model size).

Action Items:

- Shlok to prepare error analysis and confusion matrix figures.
- Sonic to format all metrics (accuracy, F1, top-5) into result tables.
- Shardul to finalize contribution summary of ensemble section.
- Sarthak to check final code reproducibility.
- Yash to refine computational efficiency and error analysis sections.

Written by: Yash Kumar Singh

Meeting 9

Date: April 13, 2025

Time: 10:15 PM – 12:15 AM

Venue: Google Meet

Attendees: All members

Discussion Points:

- Prepared and reviewed complete draft of final report.
- Verified integration of all GitHub commits:
 - Sketch prediction demo
 - Project report sections
 - Updated README, hyperparameter results
- Assigned final polishing tasks and deadlines.

Action Items:

- Shlok to finalize CNN+KNN results and visualization appendix.
- Sonic to ensure MobileNetV2 integration references are cited.
- Shardul to compile all contributions and append to report.
- Sarthak to merge all documentation and run test cases.
- Yash to edit, proofread, and format the final report PDF.

Written by: Yash Kumar Singh

Meeting 10 (Final)

Date: April 14, 2025

Time: 8:30 PM – 10:00 PM

Venue: Sarthak's Room

Attendees: All members

Discussion Points:

- Final checklist review for submission: report, codebase, visualizations, README.

- Validated ensemble implementation and final project reproducibility.
- Conducted retrospective discussion on project learnings and challenges.

Action Items:

- All members to upload final version of report and notebook on GitHub.
- Yash to submit the report and MoM document.
- Team to present ensemble pipeline in class session.

Written by: Yash Kumar Singh